

Documentation of external evaluation

Expert Review and Impact on Design

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The bottom frame to hold the side door in place while closed is fastened to the box's side with nuts and bolts. In the case that the door is open following ultrasound probing of the dragon, the user will need to close the door. In cases where the user needs the door to drop quickly (e.g. the user is in danger with the door open), the bottom track needs to stay intact. An issue can arise where the force of the door dropping applies a shear force to the fasteners holding the track to the side panel, potentially causing a fastener to break.

To combat shear-breakage of fasteners, one may attempt to distribute the force of a falling door over a greater cross-sectional area of the screws (see Figure E.1). That is, one must maximize the sum of the cross-sectional areas of these fasteners.

Two main methods exist to achieve a greater cross-sectional fastener area:

1. Increasing the quantity of the screws
2. Increasing the diameter of the shaft of the screws

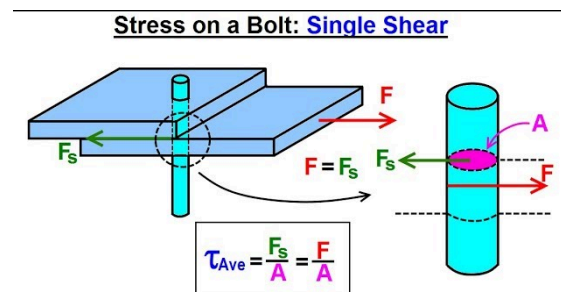
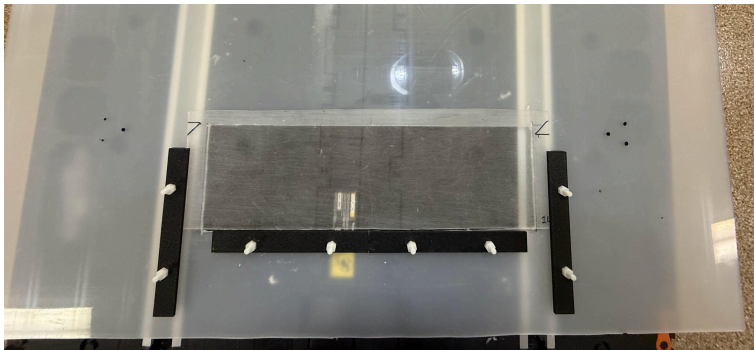


Figure E.1

Dr. Stacy Klein-Gardner, Adjunct Professor of Biomedical Engineering @ Vanderbilt University, Vanderbilt Institute for Surgery and Engineering

A latching mechanism to hold the side door must be operable by one person. Any design that requires more than two hands to open or close the side door would present a safety issue, since a user would need another individual to help close in the event of an emergency.

As a result of this advice, we implemented a ratcheting feature in our latching mechanism (shown in the figures below).



As shown in the figure, the two spring latches affixed on either side of the door assembly latch into triangular cuts in the side door.

The triangular female cutouts on the latch and door and the corresponding triangular male end of the latching spring are designed to be vertically mirrored on either side, as shown in the image below.



This design permits for ratcheting. When both springs are uncompressed and latching, the right latch prevents the door from being opened (acting as a “blocking latch”) because the blunt side of the triangle prevents the spring from compressing. Simultaneously, however, the angled side of the left triangular cutout and latch (acting as a “permitting latch”) allows upward force on the door to compresses the spring and permit door opening (see figure below).



Figure. Left latch acts as a “permitting latch.” Right latch acts as a “blocking latch.”

Conversely, in an open position, when both springs are uncompressed and latching, a user attempting to close the door would face resistance from the left latch, while the right latch will compress to allow for the door to close (an opposite outcome to the scenario above).

Two consequences are observed: (1) To open /close the door, only one hand is needed to open one latch, while another hand is used to force the door up/down. Without this feature, three hands would be needed to move in at least one of the directions: one for moving the door and two for compressing both spring latches. (2) The door can only be opened/closed when a user intentionally compresses the blocking latch.

Dr. Loudon Wright, Nashville Zoo Veterinarian